

Chapter 2

Differentiation

(微分)

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Section 2.1

The Derivative and the Tangent Line Problem

(導數與切線問題)



Tangent Line Problem (切線問題)

Let f be a real-valued function defined on $I = (a, b)$ with $c \in I$. What is the slope (斜率) m of the tangent line (切線) to the graph of f at the point $P(c, f(c))$?

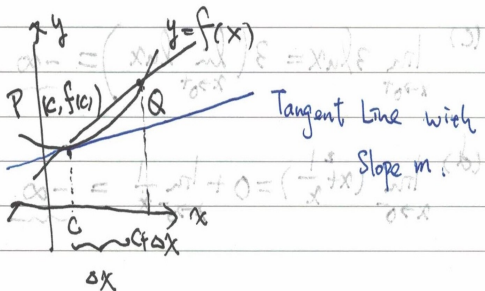


示意圖 (承上頁)

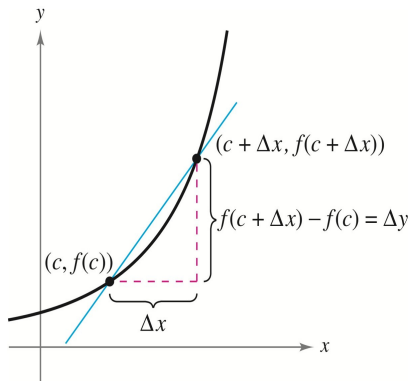
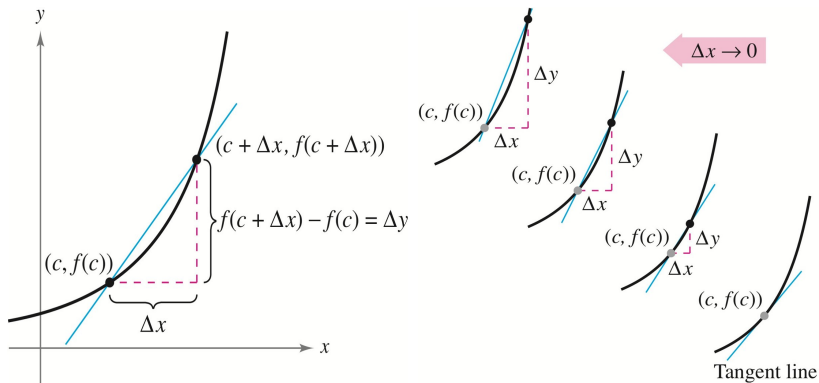


示意圖 (承上頁)



Obsevation

Since the slope of the secant line (割線) passing through $P(c, f(c))$ and $Q(c + \Delta x, f(c + \Delta x))$ is

$$m_{sec} = \frac{\Delta y}{\Delta x} = \frac{f(c + \Delta x) - f(c)}{\Delta x},$$

the slope m of the tangent line at P is determined by considering

$$m = \lim_{\Delta x \rightarrow 0} m_{sec} = \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x}.$$



Def (切線的定義)

If $m = \lim_{\Delta x \rightarrow 0} \frac{f(c+\Delta x) - f(c)}{\Delta x} = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \quad \exists$, then the line passing through $P(c, f(c))$ with slope m is called a tangent line to the graph of f at P .



Vertical Tangent Lines

Def (垂直切線的定義)

A line $x = c$ is called the vertical tangent line (垂直切線) to the graph of f at the point $(c, f(c))$ if

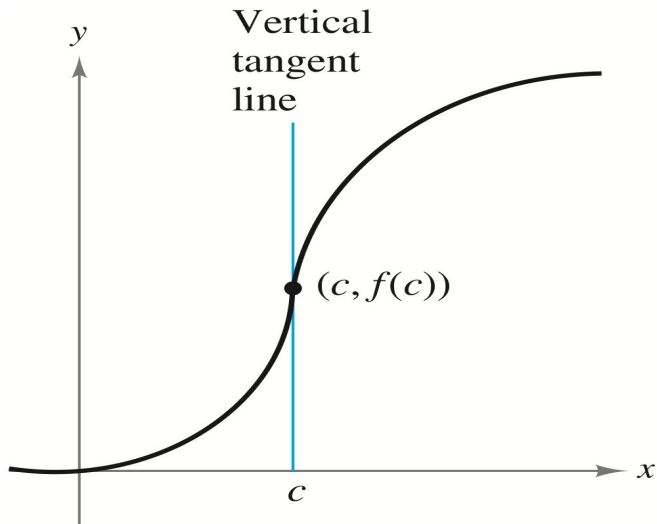
$$\lim_{\Delta x \rightarrow 0^+} \frac{f(c + \Delta x) - f(c)}{\Delta x} = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} = \pm\infty$$

or

$$\lim_{\Delta x \rightarrow 0^-} \frac{f(c + \Delta x) - f(c)}{\Delta x} = \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} = \pm\infty.$$



垂直切線的示意圖



Remark (切線方程式)

Equation of the tangent line to the graph of $y = f(x)$ at the point $(c, f(c))$ is given by

$$y - f(c) = m(x - c). \quad (\text{Point-Slope Form; 點斜式})$$



Example 4

Let $f(x) = \sqrt{x}$ for $x \geq 0$.

- (a) Find the tangent line to the graph of f at $(4, 2)$.
- (b) Find the slopes of the graph of f at $(1, 1)$ and $(0, 0)$.

Sol: For $c > 0$, the slope of a tangent line passing through $(c, f(c))$ is given by

$$\begin{aligned} m &= \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\sqrt{c + \Delta x} - \sqrt{c}}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x(\sqrt{c + \Delta x} + \sqrt{c})} = \frac{1}{\sqrt{c + 0} + \sqrt{c}} = \frac{1}{2\sqrt{c}}. \end{aligned}$$



- (a) At the point $(4, 2)$, $m = \frac{1}{2\sqrt{4}} = \frac{1}{4}$ and the equation of the tangent line at $(4, 2)$ is

$$y - 2 = \frac{1}{4}(x - 4) \quad \text{or} \quad y = \frac{1}{4}x + 1.$$

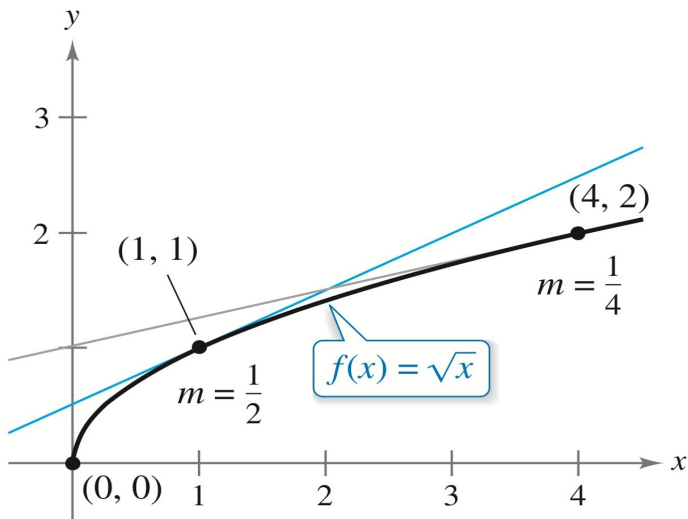
- (b) The slope of the graph of f at $(1, 1)$ is $m = \frac{1}{2\sqrt{1}} = \frac{1}{2}$. In addition, the slope of a tangent line at $(0, 0)$ is

$$\lim_{\Delta x \rightarrow 0^+} \frac{f(0 + \Delta x) - f(0)}{\Delta x} = \lim_{\Delta x \rightarrow 0^+} \frac{1}{\sqrt{\Delta x}} = \infty,$$

and hence the graph of f has a vertical tangent line at $x = 0$.



示意圖 (承上例)



Homework

Read Examples 1 and 2 by yourself.

Def (導數或導函數的定義)

(1) The derivative (導數) of f at $x \in \text{dom}(f)$ is

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}.$$

(2) f is differentiable (可微分; 簡寫為 diff.) at $x \in \text{dom}(f)$ if the derivative $f'(x)$ $\exists$.

(3) f is diff. on $I = (a, b)$ if it is diff. at each $x \in I$.



Notes

- If $S = \{x \in \text{dom}(f) \mid f'(x) \exists\}$, the first derivative f' can be regarded as a function defined on S .
- For any $y = f(x)$, the derivative is often denoted by

$$f'(x) = y'(x) = \frac{df(x)}{dx} = \frac{dy}{dx} = D_x f(x) = D_1 f(x).$$



Example 3: Find $f'(x)$ if $f(x) = x^3 + 2x$.

$$\begin{aligned}\text{Sol: } f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{(x+\Delta x)^3 + 2(x+\Delta x) - x^3 - 2x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{3x^2(\Delta x) + 3x(\Delta x)^2 + (\Delta x)^3 + 2(\Delta x)}{\Delta x} = 3x^2 + 2.\end{aligned}$$



Recall

Since the following equality

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

holds for any $a, b \in \mathbb{R}$, we immediately obtain

$$(x + \Delta x)^3 = x^3 + 3x^2(\Delta x) + 3x(\Delta x)^2 + (\Delta x)^3.$$



Thm 2.1 (可微分 $\implies$ 連續)

Let f be a real-valued function defined on $X \subseteq \mathbb{R}$ with $c \in X = \text{dom}(f)$. If f is diff. at c , then f is conti. at c .



pf: Suppose that f is diff. at $x=c$.

$$\Rightarrow f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \quad \exists$$

$$\Rightarrow \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \left[\frac{f(x) - f(c)}{x - c} (x - c) + f(c) \right]$$

$$= f'(c) \cdot 0 + f(c) = f(c). \text{ So, } f \text{ is conti. at } c.$$

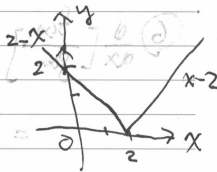


✱

Example 6 : (Thm 2.1 of $\mathbb{R}$ (3.1))

$$f(x) = |x-2| = \begin{cases} x-2, & x \geq 2 \\ 2-x, & x < 2 \end{cases} \text{ is conti.}$$

at $x=2$, but it is NOT diff. at $x=2$.



Since $\lim_{x \rightarrow 2^+} \frac{|x-2|-0}{x-2} = \lim_{x \rightarrow 2^+} \frac{x-2}{x-2} = 1$ and

$\lim_{x \rightarrow 2^-} \frac{|x-2|-0}{x-2} = \lim_{x \rightarrow 2^-} \frac{-(x-2)}{x-2} = -1$, we know that

$$f'(2) = \lim_{x \rightarrow 2} \frac{|x-2|-0}{x-2} \neq \text{exists}$$

So, f is NOT diff. at $x=2$!

HW: See Example 7. (Thm 2.1 of $\mathbb{R}$ (3.1)).



Remarks

- ① 函數的可微分點必定是連續點，詳見 Thm 2.1。
- ② 但是，函數的連續點不一定是可微分點！詳見 Example 6 和 Example 7。



Sections 2.2 & 2.3

Basic Differentiation Rules and Higher-Order Derivatives

(基本微分法則與高階導數)



Thm (Basic Differentiation Rules)

Let f and g be diff. functions of x and $c \in \mathbb{R}$. Then

$$(1) \quad \frac{d}{dx}[c] = 0.$$

$$(2) \quad \frac{d}{dx}[x^n] = n \cdot x^{n-1} \text{ for } n \in \mathbb{R}.$$

$$(3) \quad \frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x).$$

$$(4) \quad \frac{d}{dx}[c \cdot f(x)] = c \cdot f'(x).$$

$$(5) \quad \frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x) = (\text{前微})(\text{後不微}) + (\text{前不微})(\text{後微}).$$

$$(6) \quad \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2} = \frac{(\text{子微})(\text{母不微}) - (\text{子不微})(\text{母微})}{(\text{母})^2}.$$



Example 5 of Sec. 2.2 : $\frac{5-x}{5-x} \cdot \frac{0-1 \cdot 5-x}{5-x} = \frac{0-1 \cdot 5-x}{5-x}$

(a) $y = 5x^3 \Rightarrow y' = 5 \cdot 3 \cdot x^{3-1} = 15x^2$

(b) ~~$y = \frac{2}{x} = 2x^{-1} \Rightarrow y' = 2(-1)x^{-1-1} = -\frac{2}{x^2}$~~

$y = \frac{2}{x} = 2x^{-1} \Rightarrow y' = 2(-1)x^{-1-1} = -\frac{2}{x^2}$

(c) $f(t) = \frac{4}{5}t^2 \Rightarrow f'(t) = \frac{4}{5} \cdot 2 \cdot t^{2-1} = \frac{8}{5}t$

(d) $y = 2\sqrt{x} = 2x^{1/2} \Rightarrow y' = 2(\frac{1}{2})x^{1/2-1} = x^{-1/2} = \frac{1}{\sqrt{x}}$

Per-Duet



$$(e) y = \frac{1}{2^3 \sqrt{x^2}} = \left(\frac{1}{2}\right) x^{-2/3} \Rightarrow y' = \left(\frac{1}{2}\right) \left(-\frac{2}{3}\right) x^{-5/3} = \frac{-1}{3 x^{5/3}}$$

$$(f) y = \left(\frac{-3}{2}\right) x \Rightarrow y' = \frac{-3}{2}$$

Example 7, Sec 2.2 :

$$(a) f(x) = x^3 - 4x + 5 \Rightarrow f'(x) = 3x^2 - 4$$

$$(b) g(x) = \left(\frac{1}{2}\right) x^4 + 3x^3 - 2x \Rightarrow g'(x) = -2x^3 + 9x^2 - 2$$

$$(c) y = \frac{3x^2 - x + 1}{x} = 3x - 1 + x^{-1} \Rightarrow y' = 3 - x^{-2} = \frac{3x^2 - 1}{x^2}$$

HW: see Examples 2 and 4, Sec. 2.2.



Example 1 of Sec 2.3: Find $h'(x)$ if $h(x) = (3x - 2x^2)(5 + 4x)$

Sol. $h'(x) = (3 - 4x)(5 + 4x) + (3x - 2x^2)(4)$

$$= 15 - 8x - 16x^2 + 12x - 8x^2$$

$$= \underline{-24x^2 + 4x + 15}$$

Example 5 of Sec 2.3: (高法則の例3)

Find the equation of the tangent line to the graph of

$$f(x) = \frac{3 - (1/x)}{x + 5} \text{ at } (-1, 1).$$



Sol: Rewrite f as $f(x) = \frac{3x-1}{x+5} = \frac{3x-1}{x^2+5x}$

$$\Rightarrow f'(x) = \frac{3(x^2+5x) - (3x-1)(2x+5)}{(x^2+5x)^2}$$

$$= \frac{3x^2 + 15x - 6x^2 - 13x + 5}{(x^2+5x)^2}$$

$$= \frac{-3x^2 + 2x + 5}{(x^2+5x)^2}$$

For $x=-1$, we have $f'(-1)=0$

$\Rightarrow y=1$ is the tangent line to the graph of f at $(-1, 1)$.

HW: See Example 4 of Sec. 2.3.



Thm (Derivatives of Elementary Functions)

$$(1) \quad \frac{d}{dx}[\sin x] = \cos x, \quad \frac{d}{dx}[\cos x] = -\sin x.$$

$$(2) \quad \frac{d}{dx}[\tan x] = \sec^2 x, \quad \frac{d}{dx}[\cot x] = -\csc^2 x.$$

$$(3) \quad \frac{d}{dx}[\sec x] = \sec x \tan x, \quad \frac{d}{dx}[\csc x] = -\csc x \cot x.$$

$$(4) \quad \frac{d}{dx}[e^x] = e^x, \quad \frac{d}{dx}[\ln |x|] = \frac{1}{x} \text{ for } x \neq 0.$$



Sketch of the Proof

(1)

$$\begin{aligned}\frac{d}{dx}[\sin x] &= \lim_{\Delta x \rightarrow 0} \frac{\sin(x + \Delta x) - \sin x}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{\sin x \cos(\Delta x) + \cos x \sin(\Delta x) - \sin x}{\Delta x} \\&= \sin x \left(\lim_{\Delta x \rightarrow 0} \frac{\cos(\Delta x) - 1}{\Delta x} \right) + \cos x \left(\lim_{\Delta x \rightarrow 0} \frac{\sin(\Delta x)}{\Delta x} \right) \\&= (\sin x)(0) + \cos x \cdot 1 = \cos x.\end{aligned}$$

$$(2) \quad \frac{d}{dx}[\tan x] = \frac{d}{dx} \left[\frac{\sin x}{\cos x} \right] = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \sec^2 x.$$

$$(3) \quad \frac{d}{dx}[\sec x] = \frac{d}{dx} \left[\frac{1}{\cos x} \right] = \frac{(0) \cos x - 1 \cdot (-\sin x)}{\cos^2 x} = \left(\frac{1}{\cos x} \right) \left(\frac{\sin x}{\cos x} \right) = \sec x \tan x.$$



Example 2 of Sec 2.3 :

$$\frac{d}{dx}[xe^x] = 1 \cdot e^x + x \cdot e^x = e^x(x+1) \quad *$$

Example 3 of Sec 2.3 : Find y' if $y = 2x\cos x - 2\sin x$.

$$\begin{aligned} \text{Sol: } y' &= 2(x\cos x)' - 2(\sin x)' = 2(\cos x - x\sin x) - 2\cos x \\ &= -2x\sin x \quad * \end{aligned}$$

Pan.Pan



Example 8 of Sec 2.3: Find y' .

$$(a) y = x - \tan x \Rightarrow y' = 1 - \sec^2 x.$$

$$(b) y = x \sec x \Rightarrow y' = \sec x + x(\sec x \tan x) \\ = \sec x (1 + x \tan x)$$

HW: See Examples 8 and 9 of Sec. 2.2



Higher-Order Derivatives (高階導函數)

Let $y = f(x)$ be a diff. function of x . Then

- first derivative: $y' = f'(x) = \frac{dy}{dx} = \frac{d}{dx}[f(x)]$.
- second derivative: $y'' = f''(x) = \frac{d^2 y}{dx^2} = \frac{d^2}{dx^2}[f(x)] \equiv \frac{d}{dx}(y')$.
- third derivative: $y''' = f'''(x) = \frac{d^3 y}{dx^3} = \frac{d^3}{dx^3}[f(x)] \equiv \frac{d}{dx}(y'')$.
- fourth derivative: $y^{(4)} = f^{(4)}(x) = \frac{d^4 y}{dx^4} = \frac{d^4}{dx^4}[f(x)] \equiv \frac{d}{dx}(y''')$.
- $\vdots$
- n th derivative: $y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n} = \frac{d^n}{dx^n}[f(x)] \equiv \frac{d}{dx}(y^{(n-1)})$
for $n \in \mathbb{N}$. Here, we denote $y^{(0)} = y = f(x)$.



Section 2.4

The Chain Rule

(連鎖律)



Thm 2.11 (The Chain Rule; 連鎖律)

If $y = f(u)$ is a diff. function of u and $u = g(x)$ is a diff. function of x , then $y = f(g(x))$ is a diff. function of x and

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} \quad \text{or} \quad \frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x).$$



Example 3: Find $\frac{dy}{dx}$ for $y = (x^2 + 1)^3$. ✓

Sol: Let $y = f(u) = u^3$ and $u = g(x) = x^2 + 1$. Then

$$y = f(g(x)) = (x^2 + 1)^3.$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (3u^2) \cdot (2x) = 6x(x^2 + 1) \quad \text{by Chain Rule.}$$



Thm 2.12 (General Power Rule; 廣義冪法則)

If $u(x)$ is a diff. function of x , then $y = [u(x)]^n$ is also a diff. function for any $n \in \mathbb{R}$ with

$$y' = n \cdot [u(x)]^{n-1} \cdot u'(x).$$



Example 5: Find all points on the graph of $f(x) = \sqrt[3]{(x^2-1)^2}$

where $f'(x) = 0$ and $f'(x) \nexists$.

Sol: $\because f(x) = \sqrt[3]{(x^2-1)^2} = (x^2-1)^{2/3}$ with $u(x) = x^2-1$ and $n = 2/3$.

$$\Rightarrow f'(x) = \frac{2}{3}(x^2-1)^{2/3-1} \cdot (x^2-1)' = \frac{2}{3}(x^2-1)^{-1/3} (2x) = \frac{4x}{3\sqrt[3]{x^2-1}}$$

So, $f'(x) = 0$ when $x = 0$ and $f'(x) \nexists$ when $x = \pm 1$.

Per-Duet



Example 8

Find $f'(x)$ if $f(x) = \frac{x}{\sqrt[3]{x^2 + 4}} = \frac{x}{(x^2 + 4)^{1/3}}$.

Sol: From the Quotient Rule of Differentiation and the Chain Rule, we see that

$$\begin{aligned} f'(x) &= \frac{1 \cdot (x^2 + 4)^{1/3} - x \cdot (1/3)(x^2 + 4)^{-2/3}(2x)}{(x^2 + 4)^{2/3}} \\ &= \frac{1}{3}(x^2 + 4)^{-2/3} \cdot \left[\frac{3(x^2 + 4) - 2x^2}{(x^2 + 4)^{2/3}} \right] \\ &= \frac{1}{3(x^2 + 4)^{2/3}} \cdot \frac{x^2 + 12}{(x^2 + 4)^{2/3}} = \frac{x^2 + 12}{3(x^2 + 4)^{4/3}}. \end{aligned}$$



Example 11

Find the derivative y' .

(a) $y = \cos 3x^2 = \cos(3x^2)$.

(b) $y = (\cos 3)x^2$

(c) $y = \cos(3x)^2 = \cos(9x^2)$.

Sol: By the Chain Rule, we have

(a) $y' = (-\sin 3x^2) \cdot (6x) = -6x \sin(3x^2)$.

(b) $y' = (\cos 3) \cdot (2x) = (2 \cos 3)x$.

(c) $y' = (-\sin 9x^2) \cdot (18x) = -18x \sin(9x^2)$.



Example 11

$$(a) \quad y = \cos 3x^2 = \cos(3x^2) \Rightarrow y' = \frac{(-\sin 3x^2)}{3x^2} \cdot (6x) = -6x \sin(3x^2).$$

$$(b) \quad y = (\cos 3)x^2 \Rightarrow y' = 2(\cos 3)x$$

$$(c) \quad y = \cos(3x)^2 = \cos(9x^2) \Rightarrow y' = -18x \sin(9x^2).$$

$$(d) \quad y = \cos^2 x = (\cos x)^2 \Rightarrow y' = (2 \cos x)(-\sin x) = -2 \sin x \cos x.$$

$$(e) \quad y = \sqrt{\cos x} = (\cos x)^{1/2} \Rightarrow y' = \frac{1}{2}(\cos x)^{-1/2}(-\sin x) = \frac{-\sin x}{2\sqrt{\cos x}}.$$



Thm (Derivatives of e^u and $\ln |u|$)

If $u = u(x)$ is a diff. function of x , then

$$(1) \quad \frac{d}{dx}[e^x] = e^x \quad \forall x \in \mathbb{R}, \quad \frac{d}{dx}[e^u] = e^u \cdot u'.$$

$$(2) \quad \frac{d}{dx}[\ln x] = \frac{1}{x} \text{ for } x > 0, \quad \frac{d}{dx}[\ln u] = \frac{u'}{u} \text{ for } u > 0.$$

$$(3) \quad \frac{d}{dx}[\ln |x|] = \frac{1}{x} \text{ for } x \neq 0, \quad \frac{d}{dx}[\ln |u|] = \frac{u'}{u} \text{ for } u \neq 0.$$



Example 13

$$(b) \quad \frac{d}{dx} \left[\ln(x^2 + 1) \right] = \frac{u'}{u} = \frac{2x}{x^2 + 1} \text{ with } u = x^2 + 1.$$



Example 14: Differentiate $f(x) = \ln \sqrt{x+1}$ if $x > -1$.

Sol: Note that $f(x) = \ln (x+1)^{\frac{1}{2}} = \frac{1}{2} \ln (x+1)$ for $x > -1$.

$$\Rightarrow y' = f'(x) = \frac{1}{2} \frac{(x+1)'}{x+1} = \frac{1}{2(x+1)} \text{ for } \underline{x > -1}.$$



Example 15: ~~Logarithmic Differentiation~~ ~~Logarithmic Differentiation~~

Differentiate $f(x) = \ln \frac{x(x^2+1)^2}{\sqrt{2x^3-1}}$

Sol: $f(x) = \ln[x(x^2+1)^2] - \ln[(2x^3-1)^{1/2}]$
 $= \ln x + 2\ln(x^2+1) - \frac{1}{2}\ln(2x^3-1)$

$f'(x) = \frac{1}{x} + 2\left(\frac{2x}{x^2+1}\right) - \frac{1}{2}\left(\frac{6x^2}{2x^3-1}\right)$
 $= \frac{1}{x} + \frac{4x}{x^2+1} - \frac{3x^2}{2x^3-1}$

(*) HW: see Example 13 of Sec 2.4



Def (函數 a^x 和 $\log_a x$ 的定義)

Let $0 < a \neq 1$.

- (1) The exponential function to the base a (以 a 為底的指數函數) is defined by

$$a^x \equiv e^{x \ln a} \quad \forall x \in \mathbb{R}.$$

- (2) The logarithmic function to the base a (以 a 為底的對數函數) is defined by

$$\log_a x \equiv \frac{\ln x}{\ln a} \quad \forall x > 0.$$



Thm (函數 a^u 和 $\log_a |u|$ 的微分公式)

Let $u = u(x)$ be a diff. function of x and $0 < a \neq 1$.

$$(1) \quad \frac{d}{dx}[a^x] = (\ln a)a^x \quad \forall x \in \mathbb{R}, \quad \frac{d}{dx}[a^u] = (\ln a)a^u \cdot u'.$$

$$(2) \quad \frac{d}{dx}[\log_a x] = \frac{1}{(\ln a)x} \text{ for } x > 0, \quad \frac{d}{dx}[\log_a u] = \frac{u'}{(\ln a)u} \text{ for } u > 0.$$

$$(3) \quad \frac{d}{dx}[\log_a |x|] = \frac{1}{(\ln a)x} \text{ for } x \neq 0, \quad \frac{d}{dx}[\log_a |u|] = \frac{u'}{(\ln a)u} \text{ for } u \neq 0.$$



Example 16:

$$(a) y = 2^x \Rightarrow y' = (\ln 2) 2^x$$

$$(b) y = 2^{3x} \Rightarrow y' = (\ln 2) 2^{3x} \cdot (3) = 3(\ln 2) 2^{3x}$$

$$(c) y = \log_{10}(\cos x) \Rightarrow y' = \frac{-\sin x}{(\ln 10) \cos x} = \frac{-\tan x}{\ln 10}$$

$$(d) y = \log_3\left(\frac{\sqrt{x}}{x+5}\right) = \frac{1}{2} \log_3 x - \log_3(x+5)$$

$$\Rightarrow y' = \frac{1}{2(\ln 3)x} - \frac{1}{(\ln 3)(x+5)} = \frac{5-x}{2(\ln 3)x(x+5)}$$



Section 2.5

Implicit Differentiation

(隱微分)



Representations of a function

- Explicit Form (顯式):

$$y = f(x),$$

where x is the independent variable (自變量) and y is the dependent variable (應變量).

- Implicit Form (隱式):

$$F(x, y) = 0,$$

where we **assume that $y = y(x)$ is a diff. function of x .**

[Q]: How to find $y' = \frac{dy}{dx}$ under the implicit form?

Ans: apply the technique of Implicit Differentiation!



Example (補充題)

Find $\frac{dy}{dx}$ if $xy = 1$.

Sol: Assume that $y = y(x)$ is a diff. function of x . Then x and y satisfy the following equation

$$F(x, y) \equiv xy - 1 = x \cdot y(x) - 1 = 0.$$

Differentiating w.r.t. x on both sides of equality gives that

$$\frac{d}{dx}[x \cdot y(x) - 1] = y + x \cdot \frac{dy}{dx} = \frac{d}{dx}(0) = 0.$$

So, we have $\frac{dy}{dx} = \frac{-y}{x}$ for $x \neq 0$.



Example 8 (利用隱微分求切線方程式)

Find the tangent line to the graph of

$$x^2(x^2 + y^2) = y^2 \quad \text{or} \quad F(x, y) \equiv x^4 + x^2y^2 - y^2 = 0$$

at the point $(\sqrt{2}/2, \sqrt{2}/2)$.



Solution of Example 8

Sol: Assume that $y = y(x)$ is a diff. function of x . It follows from the Chain Rule that

$$4x^3 + 2xy^2 + 2x^2y \cdot \frac{dy}{dx} - 2y \cdot \frac{dy}{dx} = 0.$$

Thus, the first derivative $\frac{dy}{dx}$ is given by

$$\frac{dy}{dx} = \frac{-2x(2x^2 + y^2)}{2y(x^2 - 1)} = \frac{x(2x^2 + y^2)}{y(1 - x^2)}.$$

At the given point, $m = \left. \frac{dy}{dx} \right|_{x=y=\frac{\sqrt{2}}{2}} = \frac{3/2}{1/2} = 3$, and hence the equation of the tangent line at this point is

$$y - \frac{\sqrt{2}}{2} = 3\left(x - \frac{\sqrt{2}}{2}\right) \quad \text{or} \quad y = 3x - \sqrt{2}.$$



Example 2: Find $\frac{dy}{dx}$ given that $y^3 + y^2 - 5y - x^2 = -4$

Sol: Assume that $y = y(x)$ is a diff. function of x .

$$\frac{d}{dx} [y^3 + y^2 - 5y - x^2] = \frac{d}{dx} [-4] = 0$$

$$\Rightarrow 3y^2 \frac{dy}{dx} + 2y \frac{dy}{dx} - 5 \frac{dy}{dx} - 2x = 0 \quad \text{by Chain Rule}$$

$$\Rightarrow (3y^2 + 2y - 5) \frac{dy}{dx} = 2x \Rightarrow \frac{dy}{dx} = \frac{2x}{3y^2 + 2y - 5}$$



Example 5: Determine the slope of the graph of

$$3(x^2 + y^2)^2 = 100xy$$

at the point $(3, 1)$.

Sol: Assume that $y = y(x)$ is a diff. function of x .

$$\frac{d}{dx} [3(x^2 + y^2)^2] = \frac{d}{dx} [100xy].$$

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$$\Rightarrow 6(x^2+y^2)(2x+2y \cdot \frac{dy}{dx}) = 100y + 100x \frac{dy}{dx} \quad \text{by Chain Rule}$$

$$\Rightarrow [-100x + 12y(x^2+y^2)] \frac{dy}{dx} = 100y - 12x(x^2+y^2)$$

$$\Rightarrow \frac{dy}{dx} = \frac{100y - 12x(x^2+y^2)}{-100x + 12y(x^2+y^2)} \neq \frac{25y - 3x(x^2+y^2)}{-25x + 3y(x^2+y^2)}$$

$$\text{So, the slope at } (3,1) \text{ is } m = \left. \frac{dy}{dx} \right|_{x=3, y=1} = \frac{-65}{-45} = \frac{13}{9}$$

(*) HW: see Example 4 of Sec. 2.5



Example 7: (求 = 找 ~~求~~ 找)

Given $x^2 + y^2 = 25$, find $\frac{d^2y}{dx^2} = y''$.

Sol: Assume that $y = y(x)$ is a diff. function of x .

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25) \Rightarrow 2x + 2y \frac{dy}{dx} = 0.$$

$$\Rightarrow \frac{dy}{dx} = \frac{-x}{y}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(-1)y - (-x) \cdot \frac{dy}{dx}}{y^2} = \frac{-y + x \cdot \left(\frac{-x}{y}\right)}{y^2} = \frac{-(y^2 + x^2)}{y^3}$$

$$= \frac{-25}{y^3} \text{ because } \underline{x^2 + y^2 = 25}.$$



Logarithmic Differentiation (對數微分法)

Example 9:

Find y' if $y = \frac{(x-2)^2}{\sqrt{x^2+1}}$ for $x \neq 2$.

Sol:

$$\ln y = \ln \frac{(x-2)^2}{\sqrt{x^2+1}} = 2 \ln(x-2) - \frac{1}{2} \ln(x^2+1)$$

$$\text{By Chain Rule} \Rightarrow \frac{d}{dx} [\ln y] = \frac{d}{dx} \left[2 \ln(x-2) - \frac{1}{2} \ln(x^2+1) \right]$$

$$\Rightarrow \frac{y'}{y} = \frac{2}{x-2} - \frac{1}{2} \cdot \frac{2x}{x^2+1} = \frac{x^2+2x+2}{(x-2)(x^2+1)}$$

$$\Rightarrow y' = \frac{y(x^2+2x+2)}{(x-2)(x^2+1)} = \frac{(x-2)^2(x^2+2x+2)}{(x-2)(x^2+1)^{3/2}}$$



Section 2.6

Derivatives of Inverse Functions

(反函數的微分法則)



Thm 2.17 (反函數的可微分性)

Let f be diff. on an open interval I . If f has an inverse function $g = f^{-1}$, then g is diff. at any $x \in \text{range}(f)$ for which $f'(g(x)) \neq 0$, with the derivative

$$g'(x) = \frac{1}{f'(g(x))}.$$



Example 1: (Thm 2.17 a) (3.1.3)

$$\text{Let } f(x) = \frac{1}{4}x^3 + x - 1.$$

(a) What is $f^{-1}(3)$?

(b) What is $(f^{-1})'(3)$?



Sol:

$\because f$ is one-to-one on $\mathbb{R}$

$\therefore f^{-1} = g \exists$ on $\mathbb{R}$.

(a) Since $f(2) = 3$, $f^{-1}(3) = 2 = g(3)$.

(b) Note that $f'(x) = \frac{3}{4}x^2 + 1 \Rightarrow f'(2) = 4$.

$$\text{So, } f^{-1}(3) = g(3) = \frac{1}{f'(g(3))} = \frac{1}{f'(2)} = \frac{1}{4}$$

by Thm 2.17



Example 3: Find y' for $y = \arctan x \quad \forall x \in \mathbb{R}$.

Sol: Note that $y = \arctan x \Leftrightarrow \tan y = x$

By Chain Rule $\Rightarrow \frac{d}{dx} [\tan y] = \frac{d}{dx} (x) = 1$

$\Rightarrow \sec^2 y \cdot \frac{dy}{dx} = 1 \Rightarrow y' = \frac{1}{\sec^2 y}$ for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

$\Rightarrow y' = \frac{1}{(\sqrt{x^2+1})^2} = \frac{1}{1+x^2} \quad \forall x \in \mathbb{R}$.



$(\tan y = x)$

Thm 2.18 (6 = 31 2. 4 54 x)



Thm 2.18 (反三角函數的微分公式)

Let $u = u(x)$ be a diff. function of x . Then

$$(1) \quad \frac{d}{dx} \sin^{-1} u = \frac{u'}{\sqrt{1-u^2}}, \quad \frac{d}{dx} \cos^{-1} u = \frac{-u'}{\sqrt{1-u^2}}.$$

$$(2) \quad \frac{d}{dx} \tan^{-1} u = \frac{u'}{1+u^2}, \quad \frac{d}{dx} \cot^{-1} u = \frac{-u'}{1+u^2}.$$

$$(3) \quad \frac{d}{dx} \sec^{-1} u = \frac{u'}{|u|\sqrt{u^2-1}}, \quad \frac{d}{dx} \csc^{-1} u = \frac{-u'}{|u|\sqrt{u^2-1}}.$$



Example 4 : $(f) f = \sin^{-1}(x) \Rightarrow f' = \frac{1}{\sqrt{1-x^2}}$ (a)

$$(a) \frac{d}{dx} \sin^{-1}(2x) = \frac{2}{\sqrt{1-4x^2}}$$

$$(b) \frac{d}{dx} \tan^{-1}(3x) = \frac{3}{1+9x^2}$$

$$(c) \frac{d}{dx} \sin^{-1} \sqrt{x} = \frac{(1/2)x^{-1/2}}{\sqrt{1-x}} = \frac{1}{2\sqrt{x-x^2}}$$

$$(d) \frac{d}{dx} \sec^{-1}(e^{2x}) = \frac{2e^{2x}}{|e^{2x}| \sqrt{e^{4x}-1}} = \frac{2}{\sqrt{e^{4x}-1}}$$

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Thank you for your attention!

